

Incoming PhD student (MSU, Fall 2026) — spatiotemporal ML, deep learning weather forecasting, and AI adversarial robustness. Bridges production-grade engineering (React Native, AWS, CUDA) with advanced ML research (equivariant GNNs, MLIPs) across mobile health AI and computational materials science.

Research Experience

Research Assistant — ESC Group Jun 2024 – Present
Prof. Wenyao Xu, University at Buffalo | Buffalo, NY

- **OralScan**: Full-stack mobile health platform (React Native + web) with YOLOv8 pipelines for dental disease screening and tooth numbering; paper submitted to *Smart Health*; **2nd place \$1,000** UB Health Futures Challenge.
- **OrthoScan**: YOLO-based detection + depth measurement for remote orthodontic monitoring; reconstructed **3D dental structures** using pre-trained GNNs; computed 2D/3D tooth position changes across longitudinal visits.
- **mRehab**: Provisioned AWS backend for AI agent testing; deployed live sensor logging site; implemented **Kalman filters** for wearable IMU activity detection in live therapy workflows.

Undergraduate Researcher — Peng Lab Jun 2025 – Present
Prof. Jiayu Peng, University at Buffalo | Buffalo, NY

- **Symmetry-Aware GNNs**: Benchmarked ALIGNN on perovskite ordering energetics (arXiv:2409.13851); migrated to **Weights & Biases**; co-authored *ChemRxiv* commentary on agentic AI for catalyst discovery.
- **Equivariant MLIPs**: Implementing **E(3)-equivariant** MPNNs with alchemical graph extensions; integrating with MACE universal atom models for materials relaxation workflows.

Publications & Preprints

- Soni P., **Dandapat K.**, et al. (2025). OralScan: AI-Powered Mobile Tool for Geriatric Oral Healthcare. *Smart Health* (under review).
- Peng J., **Dandapat K.**, et al. (2025). Accelerating Multimetallic Catalyst Discovery with Agentic AI. *ChemRxiv*. doi:10.26434/chemrxiv-2025-13n3f
- Peng J., et al. (2024). Learning Ordering in Crystalline Materials with Symmetry-Aware GNNs. arXiv:2409.13851.

Selected Projects

PaddockPsychRL [GitHub] — F1-Informed Multi-Agent RL
Ray RLLib · MAPPO · PettingZoo · FastF1

- Embedded F1 temporal performance signals into per-agent psychology models; MAPPO psych-conditioned policies improved Strategy Resilience Score **0.54 → 0.63 (+16.7%)** over baselines.

Marine Guardian — Ship Detection in Satellite Imagery
TensorFlow · EfficientNet · MobileNetV2 · OpenCV

- Geometry-first pipeline (roundness classifier + **MobileNetV2**) achieved **98.72%** accuracy; benchmarked against EfficientNet and ResNet-50 transfer learning on satellite imagery.

Professional Experience

Teaching Assistant — CSE 341: Computer Architecture Jan – May 2026
University at Buffalo | Buffalo, NY

- Led weekly **Assembly & VHDL** tutorial sessions (30+ students) covering caching, pipelining, hazard detection; graded exams; supported 100+ students on Piazza.

Tutor & Peer-Assisted Leader (PAL) — TASS Aug 2024 – Dec 2025
University at Buffalo | Buffalo, NY

- PAL for STA 119: weekly worksheets + two sessions/week; ~30% quiz improvement. One-to-one tutoring: CSE 115/116/220, MTH 141/142, STA 119.

Education

PhD — Computer Science

Michigan State University
Starting Aug 2026
Advisor: **Dr. Pang-Ning Tan**
Data Mining Laboratory
Focus: spatiotemporal ML, DLWF, adversarial robustness

BS — Computer Science

Univ. at Buffalo, SUNY
Aug 2023 – Jun 2026
Minor: Statistics
GPA 3.8+ | Dean’s List (all)
Presidential Scholar \$15k/yr

Honors & Awards

PEARL Award \$2,500, Nov 2025
Health Futures Challenge 2nd place \$1,000, 2025
Cybersecurity C2C Top 100 globally; 10th in world finals, 2025
Dean’s List All semesters, UB
Presidential Scholarship \$15,000/yr

Technical Skills

Languages: Python · C++ · JS · Java · Rust
ML / AI: PyTorch · TensorFlow · YOLOv8 · GNNs · ViT · MAPPO · Ray RLLib · CUDA
Scientific ML: ALIGNN · E(3)-MPNNs · MACE · MLIPs · W&B · PyG
Engineering: React · React Native · Node.js · Django · AWS · Git · OpenCV · NumPy

Research Interests

Spatiotemporal machine learning
Deep learning weather forecasting
AI adversarial robustness
Geometric & graph deep learning
Mobile health AI
Computational materials science